

In the following exercises, round to the indicated place value.

925. Round to the nearest ten. 926. Round to the nearest hundred.
 (a) 407 (b) 8,564 (a) 25,846 (b) 25,864

In the following exercises, round each number to the nearest (a) hundred (b) thousand (c) ten thousand.

927. 864,951 928. 3,972,849

Identify Multiples and Factors

In the following exercises, use the divisibility tests to determine whether each number is divisible by 2, by 3, by 5, by 6, and by 10.

929. 168 930. 264 931. 375
 932. 750 933. 1430 934. 1080

Find Prime Factorizations and Least Common Multiples

In the following exercises, find the prime factorization.

935. 420 936. 115 937. 225
 938. 2475 939. 1560 940. 56
 941. 72 942. 168 943. 252
 944. 391

In the following exercises, find the least common multiple of the following numbers using the multiples method.

945. 6, 15 946. 60, 75

In the following exercises, find the least common multiple of the following numbers using the prime factors method.

947. 24, 30 948. 70, 84

Use the Language of Algebra

Use Variables and Algebraic Symbols

In the following exercises, translate the following from algebra to English.

949. $25 - 7$ 950. $5 \cdot 6$ 951. $45 \div 5$
 952. $x + 8$ 953. $42 \geq 27$ 954. $3n = 24$
 955. $3 \leq 20 \div 4$ 956. $a \neq 7 \cdot 4$

In the following exercises, determine if each is an expression or an equation.

957. $6 \cdot 3 + 5$ 958. $y - 8 = 32$

Simplify Expressions Using the Order of Operations

In the following exercises, simplify each expression.

959. 3^5 960. 10^8

In the following exercises, simplify

961. $6 + 10/2 + 2$ 962. $9 + 12/3 + 4$ 963. $20 \div (4 + 6) \cdot 5$
 964. $33 \div (3 + 8) \cdot 2$ 965. $(4^2 + 5^2)^2$ 966. $(4 + 5)^2$

Evaluate an Expression

In the following exercises, evaluate the following expressions.

967. $9x + 7$ when $x = 3$

968. $5x - 4$ when $x = 6$

969. x^4 when $x = 3$

970. 3^x when $x = 3$

971. $x^2 + 5x - 8$ when $x = 6$

972. $2x + 4y - 5$ when $x = 7, y = 8$

Simplify Expressions by Combining Like Terms

In the following exercises, identify the coefficient of each term.

973. $12n$

974. $9x^2$

In the following exercises, identify the like terms.

975. $3n, n^2, 12, 12p^2, 3, 3n^2$

976. $5, 18r^2, 9s, 9r, 5r^2, 5s$

In the following exercises, identify the terms in each expression.

977. $11x^2 + 3x + 6$

978. $22y^3 + y + 15$

In the following exercises, simplify the following expressions by combining like terms.

979. $17a + 9a$

980. $18z + 9z$

981. $9x + 3x + 8$

982. $8a + 5a + 9$

983. $7p + 6 + 5p - 4$

984. $8x + 7 + 4x - 5$

Translate an English Phrase to an Algebraic Expression

In the following exercises, translate the following phrases into algebraic expressions.

985. the sum of 8 and 12

986. the sum of 9 and 1

987. the difference of x and 4

988. the difference of x and 3

989. the product of 6 and y

990. the product of 9 and y

991. Adele bought a skirt and a blouse. The skirt cost \$15 more than the blouse. Let b represent the cost of the blouse. Write an expression for the cost of the skirt.

992. Marcella has 6 fewer boy cousins than girl cousins. Let g represent the number of girl cousins. Write an expression for the number of boy cousins.

Add and Subtract Integers**Use Negatives and Opposites of Integers**

In the following exercises, order each of the following pairs of numbers, using $<$ or $>$.

993.

- (a) $6 \underline{\hspace{1cm}} 2$
- (b) $-7 \underline{\hspace{1cm}} 4$
- (c) $-9 \underline{\hspace{1cm}} -1$
- (d) $9 \underline{\hspace{1cm}} -3$

994.

- (a) $-5 \underline{\hspace{1cm}} 1$
- (b) $-4 \underline{\hspace{1cm}} -9$
- (c) $6 \underline{\hspace{1cm}} 10$
- (d) $3 \underline{\hspace{1cm}} -8$

In the following exercises, find the opposite of each number.

995. (a) -8 (b) 1

996. (a) -2 (b) 6

In the following exercises, simplify.

997. $-(-19)$

998. $-(-53)$

In the following exercises, simplify.

999. $-m$ when

(a) $m = 3$

(b) $m = -3$

1000. $-p$ when

(a) $p = 6$

(b) $p = -6$

Simplify Expressions with Absolute Value

In the following exercises, simplify.

1001. (a) $|7|$ (b) $|-25|$ (c) $|0|$

1002. (a) $|5|$ (b) $|0|$ (c) $|-19|$

In the following exercises, fill in $<$, $>$, or $=$ for each of the following pairs of numbers.

1003.

(a) -8 $\underline{\hspace{1cm}}$ $|-8|$

(b) $-|-2|$ $\underline{\hspace{1cm}}$ -2

1004.

(a) $|-3|$ $\underline{\hspace{1cm}}$ $-|-3|$

(b) 4 $\underline{\hspace{1cm}}$ $-|-4|$

In the following exercises, simplify.

1005. $|8 - 4|$

1006. $|9 - 6|$

1007. $8(14 - 2|-2|)$

1008. $6(13 - 4|-2|)$

In the following exercises, evaluate.

1009. (a) $|x|$ when $x = -28$

(b) $|x|$ when $x = -15$

1010.

(a) $|y|$ when $y = -37$

(b) $|-z|$ when $z = -24$

Add Integers

In the following exercises, simplify each expression.

1011. $-200 + 65$

1012. $-150 + 45$

1013. $2 + (-8) + 6$

1014. $4 + (-9) + 7$

1015. $140 + (-75) + 67$

1016. $-32 + 24 + (-6) + 10$

Subtract Integers

In the following exercises, simplify.

1017. $9 - 3$

1018. $-5 - (-1)$

1019. (a) $15 - 6$ (b) $15 + (-6)$

1020. (a) $12 - 9$ (b) $12 + (-9)$

1021. (a) $8 - (-9)$ (b) $8 + 9$

1022. (a) $4 - (-4)$ (b) $4 + 4$

In the following exercises, simplify each expression.

1023. $10 - (-19)$

1024. $11 - (-18)$

1025. $31 - 79$

1026. $39 - 81$

1027. $-31 - 11$

1028. $-32 - 18$

1029. $-15 - (-28) + 5$

1030. $71 + (-10) - 8$

1031. $-16 - (-4 + 1) - 7$

1032. $-15 - (-6 + 4) - 3$

Multiply Integers

In the following exercises, multiply.

1033. $-5(7)$

1034. $-8(6)$

1035. $-18(-2)$

1036. $-10(-6)$